Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 30 epochs with a batch size of 128. The dataset was split into 70 percent training data and the rest for testing.

2 Results

Figure 1 shows the validation accuracy by dataset.

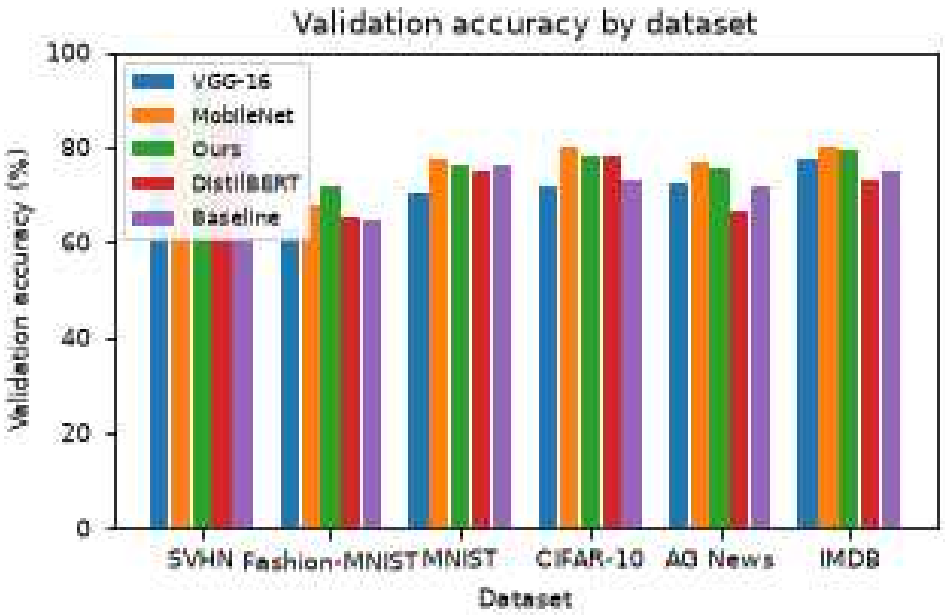


Figure 1: Validation accuracy by dataset for the models in this experiment.

As shown in Figure 1, Baseline outperforms DistilBERT by 25.1 percentage points on AG News. On AG News, MobileNet obtains a validation accuracy of 75.4%. On IMDB, MobileNet leads DistilBERT by 26.5 points.

Figure 2 shows the accuracy of Transformer by dataset.

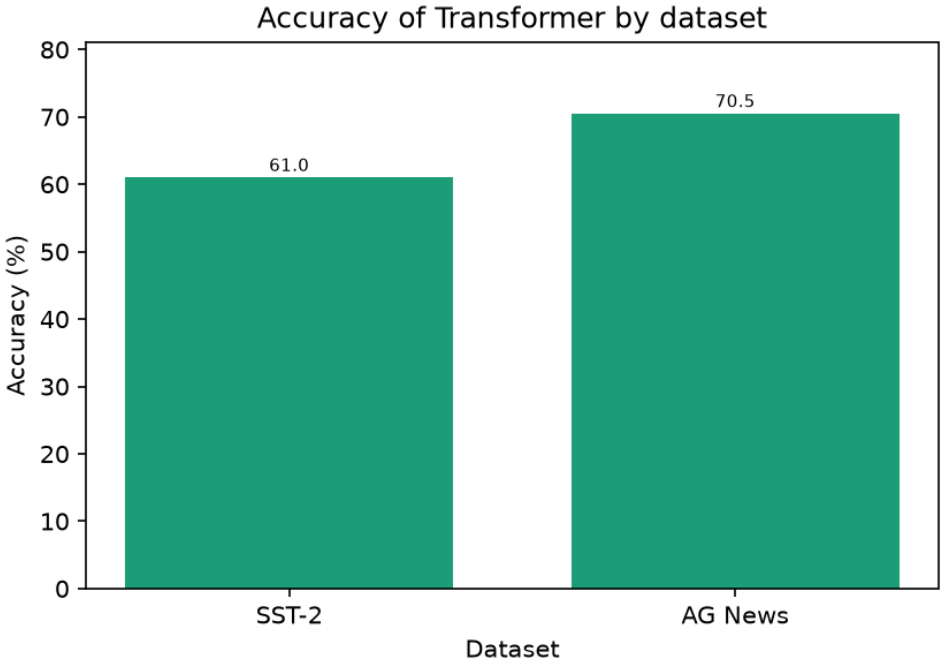


Figure 2: Accuracy of Transformer by dataset for the models in this experiment.

As shown in Figure 2, Transformer is about 10 points higher on AG News than on SST-2. Transformer scores about 10 points higher on AG News than on SST-2. Evaluated on SST-2, Transformer comes in at 61.0% accuracy.

Figure 3 shows the accuracy of Baseline by dataset.

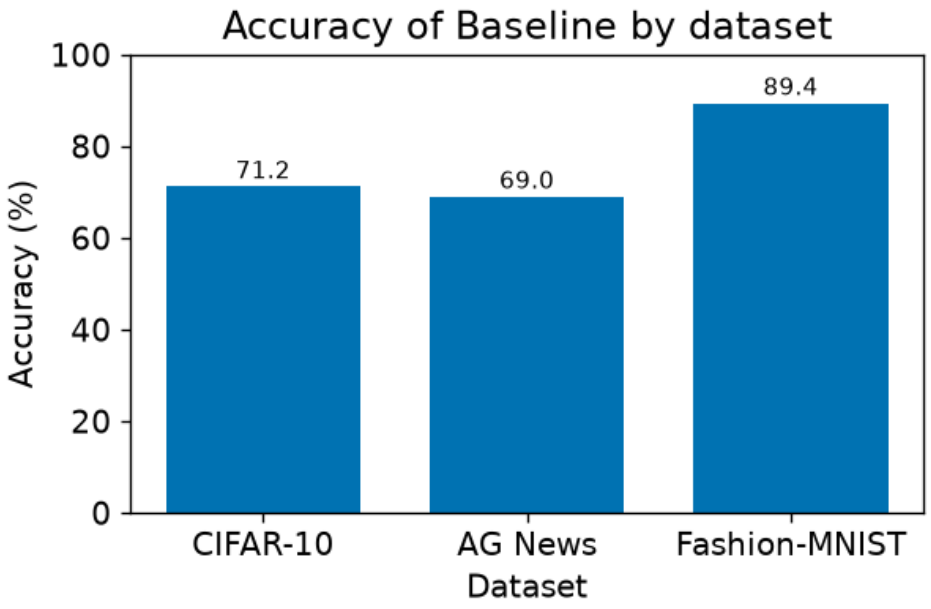


Figure 3: Accuracy of Baseline by dataset for the models in this experiment.

Baseline is 20.2 points higher on Fashion-MNIST than on CIFAR-10 (Figure 3). As shown in Figure 3, Baseline scores 7.2 points higher on CIFAR-10 than on AG News.

3 Discussion

Naive Bayes reaches 87.4% accuracy on QQP. The differences between the models are discussed in the next section.